

SuperWin v2.5.0

A Native Windows Multi-Tool Utility — WinUI 3 / C++/WinRT

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OVERVIEW

SuperWin is a single-process native Windows desktop application built with **WinUI 3 and C++/WinRT** — a code-first dashboard featuring a blended custom title bar, a **NavigationView** shell, and one dedicated page per utility. The app ships **self-contained** (the Windows App Runtime is bundled), supports **in-place automatic updates** with no admin rights required, and runs as a **single instance** with a system tray presence. It targets Windows 10/11 64-bit and currently ships **20 tools** organized across six categories. Installation is per-user with no UAC prompt; uninstall is clean via *Settings* → *Apps*.

TOOLS & FEATURES

Math & Developer

- **Unified Calculator** — Four tabs: **Class I** (basic), **Class II** (scientific with Non-Cursor TI-30Xa-style and Cursor TI-30XIIS-style modes, plus SuperMathFont v2.1 2D pretty-print toggle), **Class III** (fractions, number bases, constants), and **Class IV CAS** with full graphing engine.
- **Class IV CAS / Graphing** — TI-Nspire CX II CAS-style plotter: editable LaTeX equation list rendered live via MathLive in a WebView2 panel (tall integrals, real fraction bars, Σ/Π , radicals, roman trig). Symbolic CAS handles differentiation, integration, and simplification; numeric fallback when no closed form exists. Multi-curve plotter with per-curve color picker, show/hide, drag-to-pan, scroll/on-canvas zoom, two-tier grid, and hover coordinate trace.
- **Python IDE** — Syntax-highlighted code editor (Dev category) with one-click **Run** through any installed Python interpreter; streams program output live. Editing works without Python installed; execution requires Python on PATH.
- **JSON Formatter** — Pretty-print, minify, and validate JSON with detailed parse-error messages.
- **GUID Generator** — Random version-4 GUIDs, single or bulk, with uppercase / hyphen / brace formatting options.
- **Unit Converter** — Length, mass, temperature, data-size unit conversion, plus number base conversion.

System & Diagnostics

- **Diagnostics + Mini-Monitor** — Full hardware inventory: CPU (name, vendor, architecture, cores/threads, base clock), all GPUs + VRAM, total RAM, OS name and build (correctly reports Windows 11), device/user info, manufacturer/model, BIOS, motherboard, and every fixed drive (free/total + filesystem). Live metrics: CPU %, RAM (used/total), GPU utilization + VRAM, disk activity % + read/write throughput (PDH counters, same as Task Manager), committed/cached memory, process/thread/handle counts, and uptime. **Mini-Monitor**: always-on-top overlay with color-coded CPU/RAM/GPU/Disk bars and bold values; DPI-aware; keeps updating after navigating away from the Diagnostics page.
- **Volume Customizer** — Per-app and master volume control in exact percent and decibels via WASAPI.
- **Sleep** — Formerly Keep Awake — keeps the PC/screen awake (optionally for a fixed duration) and schedules **Restart**, **Shut Down**, or **Hibernate** in N minutes with a cancelable countdown. One-click jump to Windows Update.
- **Network Info** — Network adapters, IPv4/IPv6 addresses, MAC addresses, and quick ping.
- **Always On Top** — Global hotkey (**Ctrl+Win+T** by default) to pin or unpin any focused window above everything else, PowerToys-style.

Clipboard & Productivity

- **Clipboard** — Process-wide clipboard history captured for the entire session. Global quick-picker (**Win+Shift+V** by default): search-as-you-type, pinned items, full keyboard navigation, optional auto-paste back into the source app. Hotkey is press-to-set in Settings.
- **Notepad Super** — Rich-text editor (WinUI `RichEditBox`): font family and size, bold/italic/underline/strikethrough,

text and highlight colors (with a No Highlight clear option), alignment, bullet and numbered lists, undo/redo, RTF and TXT open/save.

- **File Converter** — Image conversion between PNG, JPG, BMP, GIF, TIFF, WEBP, HEIC, and ICO via built-in Windows Imaging (WIC). DOCX ↔ PDF conversion when Microsoft Word is installed.

Security, Media & Fun

- **Security & Privacy** — Cryptographically secure random tokens (hex/Base64), password strength meter with entropy estimate, one-click clipboard wipe.
- **Hash & Checksum** — MD5, SHA-1, and SHA-256 hashing of text or files via CNG/BCrypt.
- **Password Generator** — Configurable length and character classes with a color-coded strength readout.
- **Color Picker** — Full color wheel, hex/RGB output with copy button, and screen eyedropper.
- **Text Transform** — Case conversion, trimming, counting, and other quick text transforms.
- **Snake** — Classic Snake on a sleek themed board. Arrow keys or WASD, score tracking, restart.

ARCHITECTURE & ENGINEERING

SuperWin is a single `SuperWin.exe` process hosting a code-first WinUI 3 dashboard. An `AppHost` owns a hidden Win32 message window on the UI thread that registers the global clipboard hotkey and listens for OS clipboard changes (`AddClipboardFormatListener`), feeding a process-wide `SharedClipStore` for the whole session. The entire UI is built **code-first in C++/WinRT** with no XAML markup files, which keeps the CMake build free of the XAML compiler.

Language / UI	C++20, WinUI 3 (code-first, no XAML markup), C++/WinRT, Win32
Structure	<code>superwin_core</code> static library holds all pure logic and Win32 infrastructure (no WinUI / NuGet); WinUI pages compile only when <code>SUPERWIN_ENABLE_WINUI</code> is set. Testable logic is fully decoupled from the UI.
Build	CMake + Ninja, MSVC /W4 /permissive- /utf-8 /MP, C++20, UNICODE, NOMINMAX, WIN32_LEAN_AND_MEAN. Third-party deps via NuGet <code>PackageReference</code> : Windows App SDK, C++/WinRT, WIL, nlohmann/json, Catch2.
Auto-Update	WinHTTP appcast poller (<code>Updater.cpp</code> , no third-party dependency). On new release: prompts user, downloads portable <code>.zip</code> , swaps files in the user-writable install folder via a small PowerShell helper, relaunches. No installer UI, no admin rights.
Settings	Process-wide singleton persisted as JSON at <code>%APPDATA%\SuperWin\settings.json</code> , addressed by dotted paths (e.g. <code>clipboard.hotkey</code> , <code>ui.theme</code>).
Tests	Catch2 unit tests: <code>VolumeMath</code> , <code>ClipStore</code> , <code>HashLogic</code> , <code>ConvertLogic</code> , <code>PasswordLogic</code> , <code>TextLogic</code> . Run via <code>ctest --preset msvc-release</code> .

RELEASE & LINKS

SuperWin v2.5.0 is publicly released and actively developed. It ships as a **per-user installer** (no UAC) and a **portable ZIP** for the self-updater. The app is self-contained — no separate runtime installation required on Windows 11; the Windows App Runtime is bundled.

Landing Page	https://shankar-sachin.github.io/superwin/
Latest Release	https://github.com/shankar-sachin/superwin/releases/latest
Source	https://github.com/shankar-sachin/superwin